

RESEARCH ARTICLE

Extraction of Pure Plastic Resins From PCR Plastic Waste by Solvent-Targeted Recovery and Precipitation (STRAP)

Zhuo Xu¹ | Fei Long²  | Zach Wagner² | Charles Granger¹ | Euncheol Ra¹ | Shiying Cai² | George W. Huber¹ | Ezra Bar-Ziv²

¹Department of Chemical and Biological Engineering, University of Wisconsin – Madison, Madison, Wisconsin, USA | ²Department of Mechanical Engineering-Engineering Mechanics, Michigan Technological University, Houghton, Michigan, USA

Correspondence: Ezra Bar-Ziv (ebarziv@mtu.edu)

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ABSTRACT

We have been developing a solvent-based plastic recycling technology called STRAP. The technology is based on dissolving a targeted plastic resin in a specific solvent that does not dissolve other resins. We have demonstrated STRAP in thousands of bench scale experiments for a large variety of wastes. Recently we have demonstrated the technology for PCR, using mixed plastic wastes (MPWs), from a wet Material Recovery Facility (MRF). The process includes (1) infrared (IR) characterization to determine the plastic composition for accurate selection of the solvent to be used for the extraction of the pure resins. (2) Shredding to the right size and aspect ratio required for flowable and fast dissolvable process. (3) Mixing the MPW in the first solvent to dissolve the first resin. (4) Filtration of the solution plastic blend, to separate the nondissolved plastic from the solution. (5) Further filtration of the solution to remove micron-sized particle of pigments and fibers. (6) Cooling for precipitation. (7) Filtration of pure resins. (8) Drying of a pure resin. (9) Extrusion of the resin to pellets. (10) Generating films or other products from the pure resin. Steps 1–10 can be considered as one-cycle that extracted the first resin. (11) A second resin can be extracted with a respective solvent from the plastic that did not dissolve in the first cycle and following steps 1–10 described above. The process also includes characterization of interim and final products. The effort includes building a pilot system at 25 kg/h throughput. We will present specific results for various PCR.

1 | Introduction and Motivation

Plastic recycling has emerged as a critical topic in response to the environmental impacts arising from plastic mismanagement. Despite great efforts that have been made to address this issue, the current plastic recycling infrastructure is facing significant challenges, with the plastic recycling rate only achieving 8.7% in the United States in 2018 [1]. A primary factor contributing to these challenges is the complexity of plastic waste streams. These streams consist of a diverse array of polymers, integrated

with various additives and contaminants, which cannot be handled by conventional recycling methods.

One potential solution to mitigate the challenge above is the adoption of solvent-based technologies. We have been developing a technology named **STRAP** (Solvent-Targeted Recovery and Precipitation), which was demonstrated in thousands of bench-scale experiments, showing it can produce high-quality resins from various plastic waste streams [2–4]. In our previous studies, we have developed a method using first-principles molecular

modeling approaches to help select suitable solvents for the target polymer in the specific feedstocks, and these results can be verified by experimental data [5, 6]. With the success of the above, we would like to bring this technology to the market. We are working on designing and building a 25 kg/h pilot system.

This paper focuses on the progress we made during the past year, including (1) the midinfrared characterization techniques integrated with machine learning algorithms; (2) the preprocessing of the PCR material; and (3) the testing of the dissolution vessel and centri-sifter system.

2 | Review of Related Work

Several companies have made significant progress in commercializing solvent-based plastic recycling technologies. These companies are at various stages of development, ranging from pilot-scale operations to full commercial deployment. *PureCycle Technologies* uses a solvent-based purification process licensed from Procter & Gamble [7] to recycle polypropylene (PP) using solvents such as butane and propane at supercritical conditions. They have built a 49 000 ton/year plant that is facing various issues, and is currently operating with significantly lower capacity [8]. *APK Newcycling* is focusing on recovering LDPE and polyamides (PAs) from industrial plastic wastes [9]. They are currently operating a 8000 ton/year plant [10]. Fraunhofer Institute has developed a solvent-based recycling process named Creasolv, which has been licensed to several companies [11]. For instance, *Unilever* built a 3 ton/day plant in Indonesia that recovers PE from waset sachets. In addition, *Polystyvert* is operating a 600 ton/year plant, using ethylbenzene to extract polystyrene (PS) [12].

Despite these efforts in building solvent-based commercial plants, there are many challenges that need to be addressed

before these technologies are used more commonly in industry. The current process used in the industry can only work with certain types of plastic feedstocks with required purity, and the overall recovery efficiency would reduce when using lower-quality feedstocks [13]. Additionally, various companies are encountering challenges in producing pure resins due to the presence of diverse types of additives in the raw materials [14]. Several of the solvent plants have struggled to operate at their designed capacity. We believe that the STRAP process can solve several of the existing problems in plastic recycling.

3 | Technology Approach

We are working on developing a pilot system that can process 25 kg/h of plastic waste. Figure 1 shows its simplified process flow diagram, where MPW stands for mixed plastic wastes (MPWs). For this process, the received feedstock is shredded to increase its flowability and surface to volume ratio, which could greatly increase the dissolution efficiency. After shredding, the material is conveyed into a live-bottom hopper through a belt conveyor; the material is then brought into a dissolution vessel through a bucket elevator, with an airlock and control valve between the two pieces of equipment. After dissolution, the mixture will be drained into a heated centri-sifter, which separates the nondissolved plastics from the solution containing dissolved polymers. The nondissolved plastics from the hot filter are transferred to a solvent recovery system to recover the solvent for reuse, and the remaining solid will be stored for further use. After passing through the hot filter screen, there are still contaminants in the liquid solution such as small-sized nondissolved particles and various dissolved additives, which require additional purification. The solution will be purified to remove the remaining contaminants. The solution is then cooled in a precipitation system, which will allow the plastic to precipitate from the solvent. The

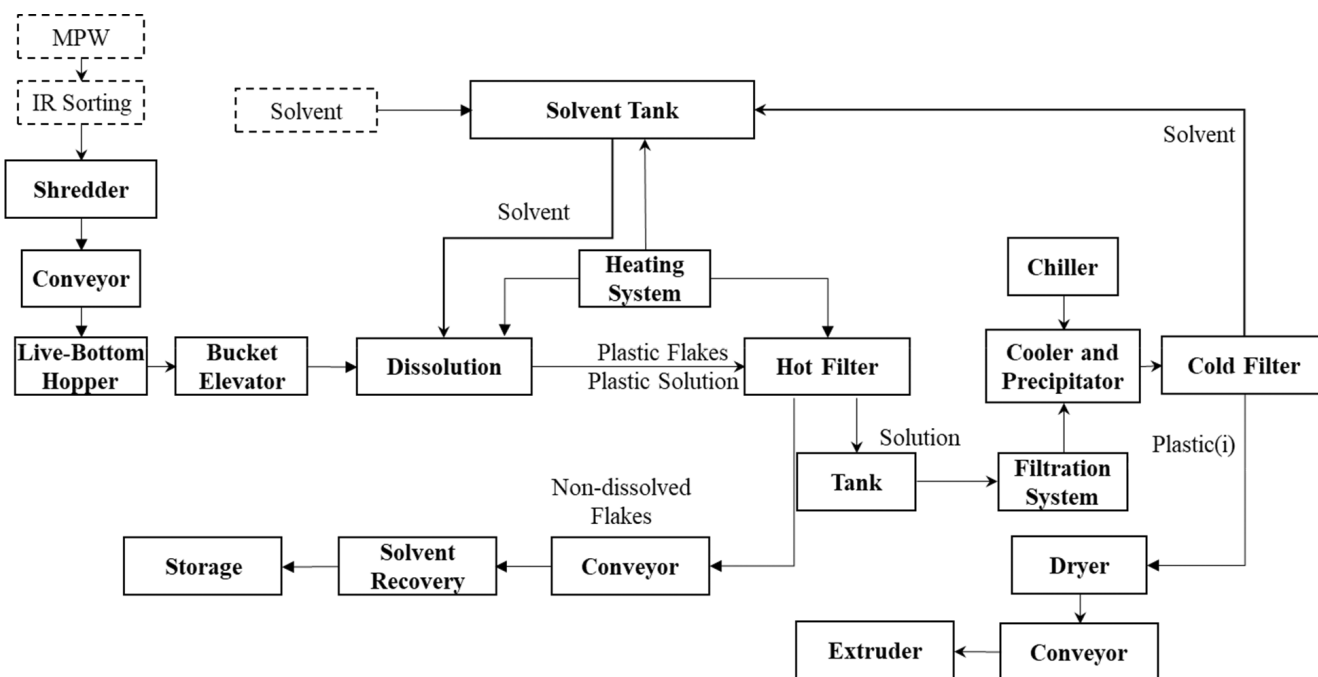


FIGURE 1 | Simplified process flow diagram of STRAP pilot system.

precipitate will then be separated by a cold filter, and the cold plastic will be dried and extruded into pellets. Compared to our previous work, we have modified the precipitation system [15]. We are in the final stages of designing the solvent drying system. Consequently, detailed descriptions of these two systems are not included in this paper.

4 | Discussion

4.1 | Real-Time MPW Characterization Using Midinfrared Spectroscopy (MIR) and Machine Learning

To effectively recycle MPWs on a large scale, many waste facilities, including municipal recycling centers and commercial recycling companies, have adopted near-infrared spectroscopy (NIR) technology as a standard practice for their operations. However, NIR suffers from limitations in detecting black plastics [16, 17], which comprise approximately 15% of MPW [18]. A promising alternative to NIR is MIR using a photon upconversion technique [19], which upconverts the midinfrared light to the near-visible region, so that it is compatible with CMOS-array detectors and can achieve a data collection rate of up to 400 spectra per second. More importantly, MIR spectroscopy can detect black plastics, which can improve the overall recycling efficiency of MPW. The experimental setup is shown in Figure 2a, and the MIR spectra of 6 different polymers measured at 400 spectra per second are shown in Figure 1b, including polycarbonate (PC), polyethylene (PE), PP, PS, and polyvinyl chloride (PVC), as well as the conveyor belt material polyurethane (PU). The shredded MPW samples are placed on a rotating platform to mimic the industrial environment where MPWs are moving on a conveyor belt. Twenty randomly selected spectra of each sample are plotted. As can be seen in Figure 1b, the MIR spectra can be obscured due to the high noise level when collected at such high speed, making the spectral identification challenging.

The team has trained a Convolutional Neural Network (CNN) model to classify the MIR spectra [20, 21]. The training data was collected at four different speeds, 400, 200, 100, and 50 spectra per second, respectively. For each sample and each speed, 5000

spectra were collected as the training dataset, and 2000 spectra were collected as the testing dataset. The confusion matrix of the CNN model is shown in Figure 3, and the classification accuracy was almost 100% for all samples. The model was validated with unlabeled MPW spectra, and the classification accuracy was also 100% for all samples, demonstrating the potential of the CNN model in the real-time characterization of MPW using MIR. As we continue working on developing this technology, it has great potential to help sort plastics from mixed waste streams with high accuracy, which could also greatly increase the efficiency of the downstream processes.

4.2 | Shredding

Downsizing of the feedstock is crucial as the received waste is usually in various sizes and shapes, which are challenging to flow. We have been collaborating with Recycle 906, a local Material Recovery Facility (MRF) that provided us with their recovered material from postconsumer wastes [22]. The material was a bale that includes various types of plastics with metal cans. The as-received material was first de-baled and the metal cans were manually removed. Figure 4a shows the material after de-baling and removal of metals, and it mainly includes plastic bottles, milk jugs, and rigid packaging materials. After the above process, the remaining plastics were granulated by a granulator (Sunly PC-400) into 10×10 mm flakes as shown in Figure 4b. According to our previous work, this material needs to be further downsized by multiple shredding [23], which not only homogenizes and downsizes the material but also creates square-shaped flakes that improve the material flow and avoid bridging. Therefore, the material was shredded multiple times by a crosscut shredder (Allegheny 16-75CX), and the final material was 2–4 mm plastic particles as shown in Figure 4c. As a result, the bulk density of the plastics increased with the number of shredding stages, as shown in Figure 5. We note that each downsizing (shredding) stage requires approximately 10 kWh/ton [4], hence, 4 stages of downsizing require about 40 kWh/ton. From experiments, we concluded that 3–4 stages are sufficient for obtaining a suitable size (2–4 mm) for fast dissolution. For example, STRAP for PE requires 600–1000 kWh/ton, which is 15–25 times larger than the downsizing energy; hence, downsizing is not a major contributor to the STRAP process. After

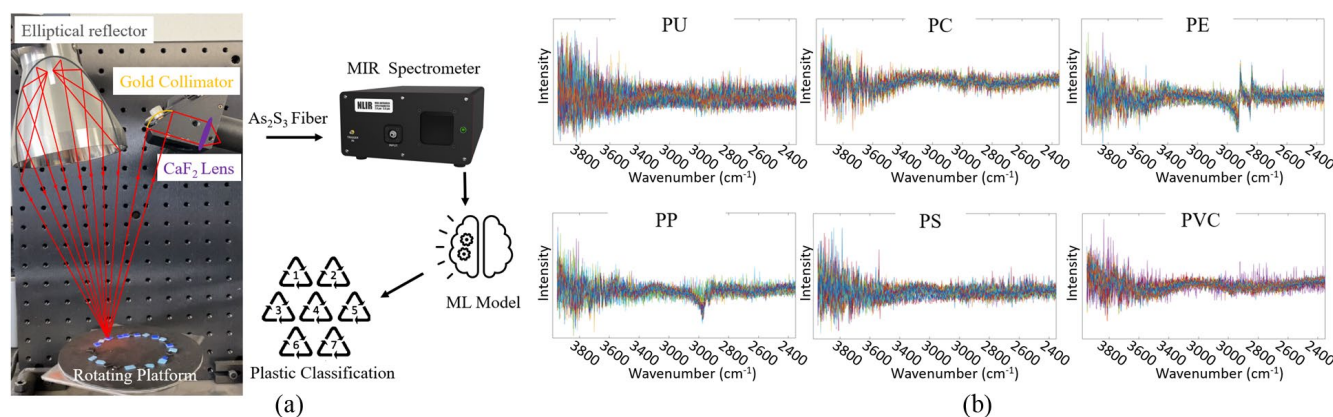
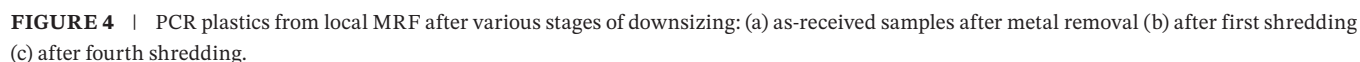


FIGURE 2 | (a) Experimental setup of real-time MPW characterization. (b) MIR spectra of various plastic samples on a rotating platform. The spectra were measured at 400 spectra per second.



reached $\sim 278 \text{ kg/m}^3$, and it gradually stabilized with additional shredding, reaching $\sim 298 \text{ kg/m}^3$.

4.3 | Dissolution Vessel

The design of the dissolution vessel has been discussed previously [15]. After the fabrication and safety testing of the vessel, we carried out extensive experiments with this vessel. We have carried out hundreds of tests using plastic and water to study the mixing and drainage of this vessel with various mixing speeds, numbers of propellers, and plastic-to-water ratios. Following extensive testing, it was discovered that the actuated ball valve located at the base of the vessel became jammed due to small plastic fragments, preventing the valve from closing properly. To address this issue, an air-actuated positioning butterfly valve was chosen as a replacement,

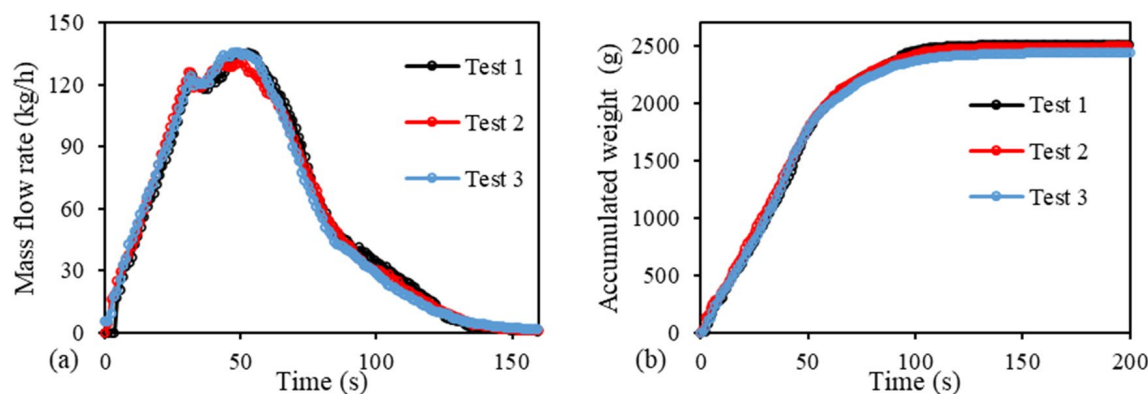


FIGURE 6 | Wet plastic weight coming out from the filter: (a) mass flow rate (b) accumulated weight.

which has been tested without further problems. Currently, we are working on building a testing hood for the dissolution and filtration system, which will allow us to operate the system with solvent.

4.4 | Centri-Sifter

The detailed configuration of the centri-sifter has also been discussed in our previous work [15]. In the current setup, we tested the unit with different screen sizes ranging from 50 to 500 μm . It quickly became apparent that the efficiency of the centri-sifter was notably diminished when utilizing screens with small sizes, and it could occasionally result in motor overheating. After extensive testing, we compared the residence time and moisture content of the final wet plastics, and the 400- μm screen had the best performance. Furthermore, during the testing phase, it was observed that the screen supplied by the original manufacturer sustained damage after prolonged and continuous testing. To address this issue, we modified the structure to reinforce the 400- μm stainless steel screen.

After opening the butterfly valve at the bottom of the dissolution vessel, the solution is drained into the centri-sifter, where the liquid passes through the screen and the solids are transferred to the solid outlet. An A&D EK-15KL scale connected to a data acquisition system is placed at the bottom of the solid outlet, which shows the weight of the material coming out in real time with a 1 s time interval. Figure 6 shows some typical results (20 kg water and 2 kg plastics) of the dynamics of the wet plastic flowing through the centri-sifter. It can be found that the wet plastics flow consistently during different tests, and all of the wet plastics come out after ~ 130 s. The mass flow rate of the wet plastics first increases after the solution is discharged, reaching ~ 140 kg/h at ~ 50 s, after which it goes down. The moisture content of the wet plastic was measured using a moisture analyzer (HFT-1000), and the average moisture content of the wet plastic after separation is 23.4% ($\pm 1.9\%$).

5 | Conclusions and Recommendations

The STRAP process has proven to be effective, at the lab scale, to recover high-quality plastic resins from feedstocks that pose challenges to the existing recycling infrastructure. This paper

summarizes the recent progress of the development of the STRAP pilot-scale system. It shows that: (1) MIR integrated with machine learning algorithms can help identify the plastic types with high accuracy and efficiency, which will benefit the downstream process to determine the process parameters; (2) shredding with a low-shear crosscut shredder can reduce the plastic particle sizes and increase its bulk density, which will increase the flowability of the feedstock and increase the dissolution efficiency; (3) a ball valve is inadequate for managing the discharge of the solution, while the butterfly valve proved capable of performing this task; and (4) the centri-sifter with a modified screen can consistently separate the plastic from water with high efficiency, and for 20 kg water and 2 kg plastic, the final moisture content of the wet plastic is 23.4% ($\pm 1.9\%$).

However, there still exist certain knowledge gaps in the ongoing development of the STRAP pilot system: (1) removal of additives and other contaminants from the plastic to produce high-purity resin; (2) the impurity buildup in the reused solvent; (3) product qualities required by the customer; (4) comprehensive technoeconomic assessment and lifecycle assessment. STRAP requires extensive work in the understanding of the interaction mechanisms of solvent-polymer and solvent-additive, along with enhanced collaborations with industrial partners.

Author Contributions

Zhuo Xu did the sample processing, dissolution vessel and centri sifter design, and write the manuscript. Fei Long developed the MPW characterization system, and revised the manuscript. Zach Wagner tested the dissolution vessel and centri sifter. Charles Granger did lab-scale experiments to guide the design of pilot-scale STRAP system. Euncheol Ra did lab-scale experiments to guide the design of pilot-scale STRAP system. Shiyong Cai did sample preparation and preprocessing. George W. Huber developed the idea of STRAP, and revised the manuscript. Ezra Bar-Ziv led the pilot-scale STRAP system development, and revised the manuscript.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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